

E-Commerce Customer Churn Analysis: Understanding Why Users Leave

Portfolio Case Study | Customer Retention Project

Project Background & Objective

I wanted to dig into why online stores lose customers, so I analyzed a dataset of **50,000 shoppers**. The main metric that stood out right away was a baseline **churn rate of 28.9%**. That means nearly one out of every three customers was leaving the platform, which is a major leaky bucket for any e-commerce business.

The core objective of this analysis was to look past the high-level percentages, identify exactly where users are hitting roadblocks in their lifecycle, and map out straightforward, actionable product improvements to protect recurring customer revenue.

What the Data Revealed

Looking closely at the user metrics, it became obvious that customer churn isn't random. It is almost entirely concentrated around two specific operational friction points:

- **Customer Service Volume:** Customers who are happy and stay with the platform rarely call support. However, customers who end up leaving show an aggressive spike in support calls, averaging between **6.4 and 6.9 calls per user**. This tells us there's a serious early-stage usability issue forcing them to constantly seek help before they finally give up.
- **Cart Abandonment:** There is a direct, heavy overlap between high cart abandonment rates and user churn. Shoppers are actively finding products and adding them to their carts, but they are hitting a wall during the final checkout steps and walking away at the last second.

Customer Tier	Observed Behaviors	Retention Risk Level
High Friction	Call support frequently (5+ times), high drop-offs at checkout	Critical Risk - Primary cause of customer churn
Neutral	Normal browsing, occasional support interaction, cyclical purchases	Moderate Risk - Vulnerable to leaving if friction increases
Low Friction	Rarely call support, seamless checkouts, consistent buying patterns	Low Risk - Highly loyal, core recurring buyers

Opportunities for Improvement

Based on these two big friction points, I formulated two straightforward product improvements that could realistically fix these leaks:

1. Simplifying the First 14 Days (Fixing Support Friction)

Since high support calls directly predict churn, we need to solve customer confusion before they feel forced to pick up the phone. Introducing a simple, guided interactive onboarding walkthrough during a new user's first two weeks would help clarify common setup steps and lower that early friction.

2. Fixing Checkout Drop-offs (Fixing Cart Abandonment)

Because users are abandoning full carts, the platform is losing high-intent revenue at the very last second. We should test a simplified, one-click checkout option or introduce automated, live-chat assistance prompts if a customer gets stuck on the payment page for more than 45 seconds.

Measuring Success

To test these ideas safely, they should be launched as small A/B tests against a random group of users. I would track success using a simple metric hierarchy:

- **The Main Goal:** Bring the overall churn rate down from 28.9% to a target of 24.5%.
- **The Core Indicators:** Track whether the average number of support calls drops and whether cart completion rates go up for the users who get the new features.
- **The Safety Check:** Keep an eye on Average Order Value (AOV) to ensure that simplifying the checkout process doesn't accidentally cause people to spend less money overall.